

What Drains Your Battery? A Continuous-Time, Physics-Compliant Model for Smartphone Battery Discharge Prediction

Abstract

Predicting the Time-to-Empty (TTE) of smartphones is complicated by the nonlinear electrochemical dynamics of Lithium-ion batteries and stochastic user workloads. This paper establishes a robust **Second-Order Resistance-Capacitance (2nd-Order RC) Equivalent Circuit Model coupled with a dynamic power consumption framework** to bridge the gap between theoretical capacity and real-world endurance.

First, we construct a continuous-time differential equation system that integrates a 7th-order polynomial Open Circuit Voltage (OCV) function with dual polarization branches. Unlike static resistance models, our framework rigorously solves the algebraic loop between terminal voltage and current under constant-power constraints ($P = V \cdot I$). This approach accurately captures the “voltage collapse” phenomenon—where terminal voltage hits the cutoff threshold before SOC depletion due to significant ohmic (IR_0) and polarization voltage drops during high-power tasks.

Second, we validate the model across diverse realistic scenarios, including Gaming, Streaming, and Idle states. The simulation reveals that TTE is not merely a function of capacity but is governed by a regime-dependent nonlinearity: high-power loads trigger accelerated voltage decay.

Finally, translating these physical insights into actionable strategies, we propose a **Confidence-Based Power Management System**. We recommend that Operating Systems (OS) replace mean-based TTE estimates with conservative 90% confidence intervals to prevent unexpected shutdowns. Furthermore, we suggest a **Preemptive Voltage Governor** that throttles CPU frequency when terminal voltage approaches the cutoff, actively recovering voltage potential to extend emergency runtime. For users, our analysis confirms that optimizing low-power background tasks yields a significantly higher marginal gain in battery duration than limiting active high-performance usage.

Keywords: Second-Order RC Model; Constant-Power Discharge; Time-to-Empty (TTE); Voltage Collapse; Power Management Strategy.

Contents

1	Introduction	3
1.1	Research Background	3
1.2	Problem Statement	3
2	Model Assumptions and Nomenclature	4
2.1	Fundamental Assumptions	4
2.2	Nomenclature	6
3	Problem 1: Model Development and Solution	7
3.1	Model Development	7
3.1.1	Circuit Architecture and Physical Mechanisms	8
3.1.2	Characterization and Fitting of Open-Circuit Voltage	9
3.1.3	System Dynamic Equations	9
3.2	Model Solution	10
3.2.1	Algebraic Loop Elimination	10
3.2.2	Pure Differential Equation System	11
3.2.3	Numerical Solution Algorithm	11
3.3	Power Consumption Model	12
3.4	Results and Analysis	13
3.4.1	Battery Discharge Characteristics in Typical Usage Scenarios	13
3.4.2	Battery Performance Under a Realistic Daily Usage Pattern	14
3.4.3	Comparative Analysis of Power Consumption Across Scenarios	15
4	Problem 2: Discharge Time Prediction Under Realistic Conditions	16
4.1	Problem Restatement and Mathematical Formulation	16
4.2	Numerical Solution and Dynamic Analysis	17
4.2.1	Voltage Collapse Under High Load	17
4.3	Comprehensive Discharge Time Prediction Landscape	17
4.3.1	Analysis of the TTE Prediction Landscape	18
5	Problem 3: Sensitivity Analysis and Robustness Assessment	19
5.1	Non-linear Impact of Power Load	19
5.2	Capacity-Power Coupling Trade-off	20
5.3	Uncertainty Quantification via Monte Carlo Simulation	21
6	Problem 4: Practical Implementation and Model Extensions	24
6.1	User Guidelines: Exploiting the High-Sensitivity Region	24
6.2	OS Strategy: Probabilistic Power Management	24
6.3	Model Extension: Analytical Derivation for Temperature and Aging	24
	References	26

1 Introduction

1.1 Research Background

As the central enabler of modern information society, the performance of smartphones is intrinsically linked to the precise management of their energy system—the lithium-ion battery. Accurate and reliable estimation of the State of Charge (SOC) constitutes a critical technical challenge for ensuring battery safety, optimizing energy management strategies, and enhancing user experience. Current research on battery modeling and state estimation primarily follows two distinct technical pathways: one is based on physical electrochemical models¹, which offer high descriptive accuracy but are computationally intensive; the other relies on empirical equivalent circuit models^{2,3}. These models simulate the macroscopic dynamic response of batteries by constructing electrical networks composed of lumped elements such as voltage sources, resistors, and capacitors, and are frequently combined with Kalman filter-based algorithms⁴ for state estimation. Owing to their clear physical interpretation and moderate computational complexity, equivalent circuit models have become the most widely adopted solution in practical Battery Management Systems (BMS).

To further enhance model accuracy and algorithmic robustness under complex operating conditions, recent research has focused on deepening the exploration of the physical underpinnings and application boundaries of equivalent circuit models:

Venegas et al.⁵ introduced the hysteresis effect of the open-circuit voltage into the model, aligning the dynamic response of the polarization process more closely with the nonlinear behavior of actual batteries. This work provides a significant reference for refining polarization voltage modeling in second-order RC models.

Mu et al.⁶ employed fractional calculus theory to characterize the relaxation processes of batteries, expanding the descriptive capacity of equivalent circuit models for complex battery dynamics. Their approach offers a novel perspective for optimizing the identification of time constants in polarization elements.

Xiong et al.⁹ enhanced the adaptability of equivalent circuit models to battery aging by integrating capacity degradation models, demonstrating the feasibility and effectiveness of incorporating State of Health (SOH) corrections within the equivalent circuit framework.

In summary, state estimation methods based on equivalent circuit models have demonstrated considerable potential for addressing complex real-world scenarios. These recent advancements lay a solid theoretical and technical foundation for the present study, which aims to construct a continuous-time "Second-Order RC Equivalent Circuit" model that incorporates the coupled dynamics of power consumption.

1.2 Problem Statement

The operational power consumption of smartphones exhibits highly dynamic and uncertain characteristics, influenced by the coupled effects of multiple factors such as screen

brightness, Central Processing Unit (CPU) load, wireless communication module activity, background applications, and ambient temperature. Against this backdrop, constructing a mathematical model that can precisely describe the continuous-time evolution of battery SOC under realistic usage conditions, and subsequently enabling reliable prediction of the remaining battery depletion time across various typical usage scenarios, emerges as a critical problem to be solved.

In accordance with the stated requirements, this study is tasked with addressing the following specific problems:

1. **Problem 1:** Establish a dynamic lithium-ion battery model based on continuous-time differential equations. This model must accurately characterize evolution of battery SOC with other factors.
2. **Problem 2:** Calculate or approximate the depletion time under various initial SOC levels and usage scenarios based on the model. Compare the prediction results with observed or reasonable behaviors, quantify the associated uncertainties, and identify conditions under which the model performs well or poorly.
3. **Problem 3:** Systematically evaluate the impact of key model parameters and different usage patterns (e.g., fluctuating power levels) on the predicted battery depletion time, thereby quantifying the sources of model uncertainty.
4. **Problem 4:** Translate your findings into practical recommendations for a cellphone user and consider how battery aging reduces effective capacity or how your modeling framework could generalize to other portable devices.

This study will proceed by addressing the objectives outlined above, striving to establish an effective bridge between theoretical modeling and engineering application.

2 Model Assumptions and Nomenclature

2.1 Fundamental Assumptions

To develop a mathematically tractable model that effectively captures the key dynamics of the battery, this study introduces the following fundamental assumptions in establishing the second-order RC equivalent circuit model, based on the physical characteristics of lithium-ion batteries and typical application scenarios:

The aforementioned assumptions aim to strike a balance between model complexity and practicality. Assumptions 1 and 2 form the physical basis of the second-order RC equivalent circuit, simplifying the distributed-parameter, nonlinear electrochemical system into a lumped-parameter, linear dynamic system. Assumption 3 defines the core operational condition (constant-power discharge) and initial/boundary conditions for this study. Assumption 4 significantly simplifies the parameter identification and model-solving process, which is common in engineering practice. Under this combined framework, the dynamic equations are established and can be solved efficiently using numerical methods. All subsequent simulations and analyses are conducted within this framework.

Table 1: Fundamental Model Assumptions

Category	Description
1. Circuit Equivalence	The complex internal electrochemical processes (e.g., lithium-ion intercalation/deintercalation, solid-phase diffusion, charge transfer) can be sufficiently approximated by a linear circuit network composed of lumped-parameter elements (resistors, capacitors, voltage sources). Specifically, a topology consisting of an open-circuit voltage source in series with an ohmic resistor and two parallel RC branches is adopted.
2. Electrochemical Simplifications	a) The internal temperature of the battery is assumed uniform and constant, neglecting dynamic effects of temperature variation on model parameters. b) The open-circuit voltage V_{oc} is solely a single-valued function of SOC, and its hysteresis effect is considered negligible. c) The polarization dynamics can be described by first-order linear kinetics, with time constants remaining constant during a single discharge process.
3. Operational & Boundary Conditions	a) The primary focus is on constant-power (P_0) discharge conditions, where the output power is precisely set by the external load and remains constant. b) Discharge starts from a fully relaxed battery state, i.e., with zero initial polarization voltages: $V_{p1}(0) = V_{p2}(0) = 0$. c) The Coulombic efficiency η is constant during discharge, typically assumed to be 1. d) The power conversion efficiency from the battery terminals to the load, η_{conv}, is constant.
4. Parameter Properties	a) For a single simulation, the polarization resistances R_1, R_2 and capacitances C_1, C_2 are treated as constants, independent of SOC, current, or temperature. Their values are taken from the open dataset from the University of Maryland¹⁶ b) The nominal battery capacity C_n and the State of Health (SOH) are known constants. c) The battery terminal voltage must not fall below a preset cutoff voltage V_{min}; discharge is terminated when this limit is reached.

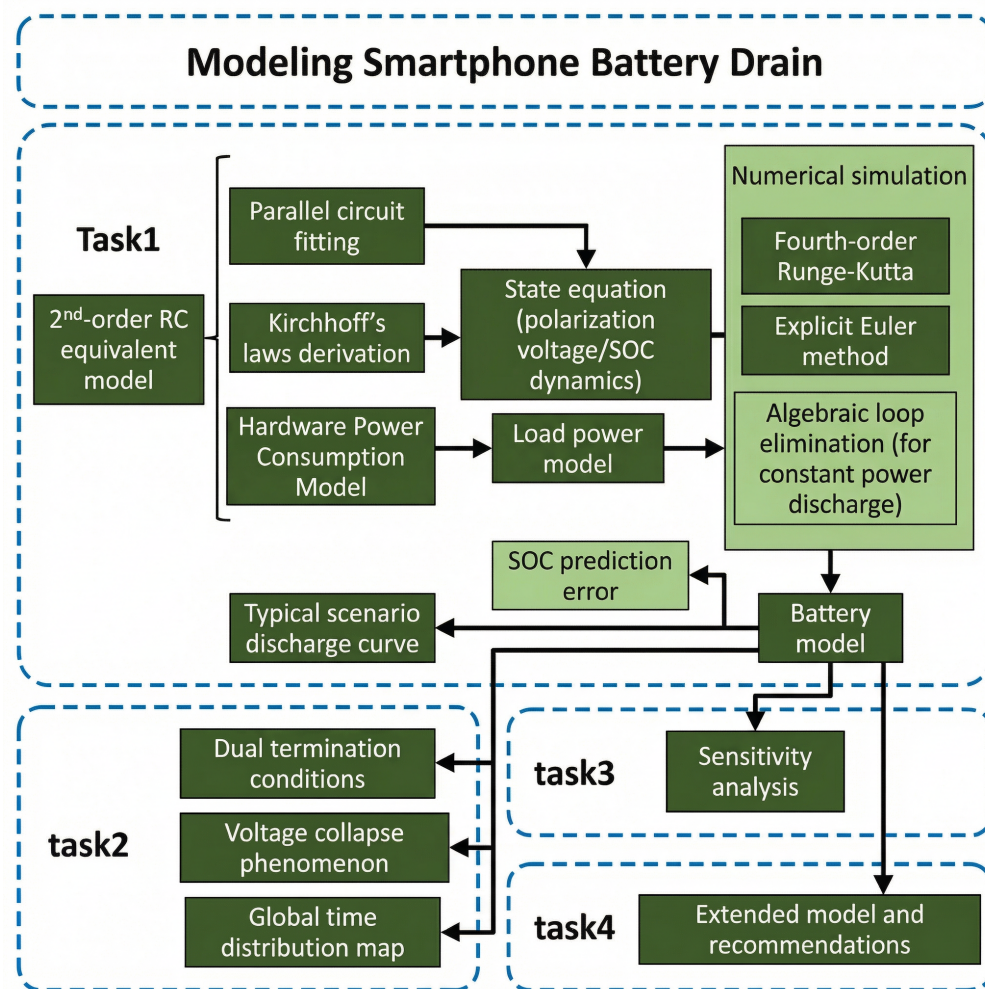


Figure 1: Framework flowchart of smartphone battery discharge modeling

2.2 Nomenclature

The main symbols used in this paper and their descriptions are listed in Table 2.

Table 2: Nomenclature

Symbol	Description	Unit
V_{oc}	Battery open-circuit voltage, function of SOC	V
SOC	State of Charge, $0 \leq \text{SOC} \leq 1$	–
SOH	State of Health, indicating capacity fade, $0 < \text{SOH} \leq 1$	–
V_t	Battery terminal voltage	V
I	Battery current (positive for discharge)	A
P_0	Constant discharge power	W
R_0	Ohmic internal resistance	Ω
R_1, C_1	Polarization resistance and capacitance of the 1st RC network (describing electrochemical polarization)	Ω, F
R_2, C_2	Polarization resistance and capacitance of the 2nd RC network (describing concentration polarization)	Ω, F
V_{p1}	Polarization voltage across the 1st RC network	V
V_{p2}	Polarization voltage across the 2nd RC network	V
$\tau_1 = R_1 C_1$	Time constant of the 1st RC network	s
$\tau_2 = R_2 C_2$	Time constant of the 2nd RC network	s
C_n	Nominal battery capacity	Ah
η	Coulombic efficiency (typically 1 for discharge)	–
η_{conv}	Power conversion efficiency	–
$V_{\min}$	Discharge cutoff voltage	V
$\text{SOC}_{\min}$	Minimum allowable SOC	–
t	Time	s
Δt	Numerical integration time step	s
$\mathbf{x} = [\text{SOC}, V_{p1}, V_{p2}]^\top$	System state vector	–
$P_{\max}$	Instantaneous maximum deliverable power	W
Q_{disch}	Cumulative discharged capacity	Ah
I_{avg}	Average discharge current	A
t_{end}	Discharge termination time	s

3 Problem 1: Model Development and Solution

3.1 Model Development

To accomplish the objective of Task 1, an equivalent circuit model was first developed in this study. This model abstracts the dynamic behavior of the battery into lumped-parameter circuit components, thereby capturing the coupled evolution of the State of Charge (SOC), terminal voltage, and polarization state under constant-power discharge conditions.

3.1.1 Circuit Architecture and Physical Mechanisms

To characterize the dynamic characteristics of lithium-ion batteries, a second-order RC equivalent circuit model is adopted. This model is constructed based on the internal electrochemical processes, abstracting and representing the complex physical mechanisms as a network of lumped-parameter circuit elements. It achieves a favorable trade-off between modeling accuracy and computational burden.

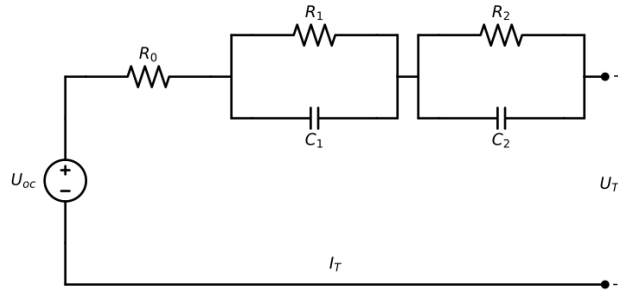


Figure 2: The Second-Order RC Equivalent Circuit Model for Lithium-Ion Batteries

As illustrated in Figure 2, the second-order RC equivalent circuit model(2ndRC-ECM) is primarily composed of the following key components connected in series:

- (a) **Open-Circuit Voltage Source** $V_{oc}(\text{SOC})$: It is used to represent the thermodynamic equilibrium potential of the battery. Its magnitude is a monotonically nonlinear function of the state of charge (SOC). To accurately capture this relationship, a seventh-order polynomial is employed for the fitting, as given by Equation:

$$V_{oc}(\text{SOC}) = \sum_{k=0}^7 a_k \text{SOC}^k \quad (1)$$

- (b) **Ohmic Internal Resistance** $R_0(\text{SOC})$: To simplify the model, this study initially assumes a constant temperature. Under this condition, the relationship between R_0 and SOC is constructed using an interpolation function based on experimental data from¹⁸, as follows:

$$R_0(\text{SOC}) = \text{interp}(\text{SOC}, \{\text{SOC}_i\}, \{R_{0,i}\}) \quad (2)$$

where $\{\text{SOC}_i\}$ and $\{R_{0,i}\}$ represent the discrete SOC points and corresponding ohmic resistance values obtained from experimental measurements, and $\text{interp}(\cdot)$ denotes a cubic spline interpolation function to ensure smooth and continuous variation of resistance with SOC.

- (c) **First RC Network** (R_1, C_1): Simulates the **electrochemical polarization** process, corresponding to the transfer impedance and double-layer capacitance effects at the electrode/electrolyte interface. Its time constant $\tau_1 = R_1 C_1$ is typically in the order of seconds, reflecting the fast dynamic behavior of the system.

- (d) **Second RC Network** (R_2, C_2): Simulates the **spontaneous polarization** process, corresponding to the solid diffusion of lithium ions within the active material particles of the electrode. Its time constant $\tau_2 = R_2 C_2$ is typically in the order of hundreds of seconds, reflecting the slow dynamic behavior of the system.

3.1.2 Characterization and Fitting of Open-Circuit Voltage

Leveraging the open-source experimental datasets from the CALCE laboratory at the University of Maryland¹⁶, a high-order polynomial regression is employed to model the OCV-SOC relationship. Specifically, analysis of the standard charge-discharge cycle data determined that a seventh-order polynomial can accurately capture the nonlinearity of this relationship, with a coefficient of determination (R^2) exceeding 0.99. The resulting fitted expression is given in Equation 3:

$$V(\text{SOC}) = \sum_{k=0}^7 a_k \cdot \text{SOC}^k \quad (3)$$

$$a_7 = 66.939, a_6 = -248.739, a_5 = 371.911, a_4 = -291.517, \\ a_3 = 132.659, a_2 = -36.387, a_1 = 6.069, a_0 = 3.259$$

To visually assess the fitting performance, Figure 3 presents a comparison between the experimentally measured OCV-SOC data points and the seventh-order polynomial fitted curve. As shown, the fitted curve smoothly traverses the data points and exhibits strong agreement across the entire SOC range (0-1), thereby validating the high accuracy and reliability of the fitted relationship.

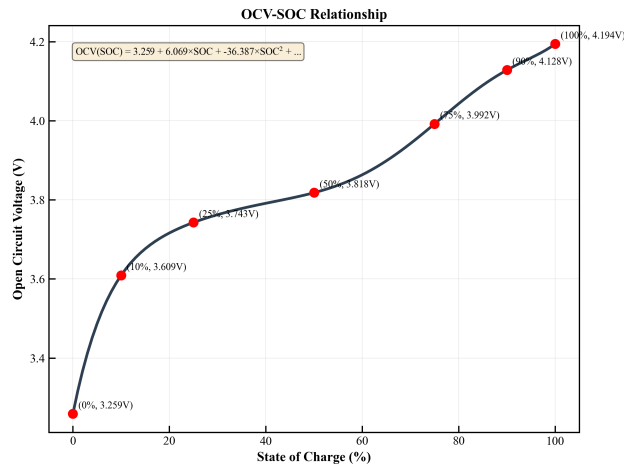


Figure 3: Measured Data and Polynomial Fitted Curve for V_{oc} vs. SOC Relationship

3.1.3 System Dynamic Equations

Based on Kirchhoff's Voltage Law (KVL) and Current Law (KCL), the continuous-time state-space equations of the model can be established.

Voltage Polarization Dynamics For the first RC network, the current I_{C_1} through the capacitor C_1 satisfies $C_1 dV_{p1}/dt = I_{C_1}$. The current $I_{R_1} = V_{p1}/R_1$ through the resistor R_1 . According to KCL, $I(t) = I_{C_1} + I_{R_1}$. Therefore, we have:

$$\frac{dV_{p1}}{dt} = \frac{I(t)}{C_1} - \frac{V_{p1}}{R_1 C_1} \quad (4)$$

Similarly, for the second RC network:

$$\frac{dV_{p2}}{dt} = \frac{I(t)}{C_2} - \frac{V_{p2}}{R_2 C_2} \quad (5)$$

State of Charge (SOC) Dynamics SOC is defined as the ratio of the current remaining capacity to the nominal capacity. Its change rate is directly proportional to the current:

$$\frac{dSOC}{dt} = -\frac{\eta I(t)}{C_n \cdot SOH} \quad (6)$$

where η denotes the Coulombic efficiency, typically set to $\eta = 1$ during discharge, implying ideal conditions without side reactions; C_n is the nominal battery capacity (Ah); and SOH represents the State of Health, defined as the ratio of the current effective capacity to the nominal capacity, quantifying the degree of capacity fade.

Output Voltage Equation The terminal voltage $V_t(t)$ of the battery is the open-circuit voltage minus all internal voltage drops:

$$V_t(t) = V_{oc}(SOC) - I(t)R_0(SOC) - V_{p1}(t) - V_{p2}(t) \quad (7)$$

Power Constraint Under the preset conditions, we assume that the battery is discharged at a constant power P_0 , with the current $I(t)$ and terminal voltage $V_t(t)$ satisfying:

$$I(t) = \frac{P_0}{\eta_{conv} V_t(t)} \quad (8)$$

where η_{conv} denotes the power conversion efficiency. Equations (7) and (8) form a closed loop, resulting in a differential-algebraic equation system (DAE).

3.2 Model Solution

3.2.1 Algebraic Loop Elimination

To solve the DAE system numerically, we need to eliminate the algebraic loop. Substituting Equation (8) into Equation (7) gives:

$$V_t = V_{oc}(SOC) - \frac{P_0 R_0(SOC)}{\eta_{conv} V_t} - V_{p1} - V_{p2} \quad (9)$$

After rearranging, we obtain a quadratic equation in V_t :

$$V_t^2 - [V_{oc}(SOC) - V_{p1} - V_{p2}] V_t + \frac{P_0 R_0(SOC)}{\eta_{conv}} = 0 \quad (10)$$

The solutions of Equation (10) are:

$$V_t = \frac{[V_{oc}(SOC) - V_{p1} - V_{p2}] \pm \sqrt{[V_{oc}(SOC) - V_{p1} - V_{p2}]^2 - \frac{4P_0 R_0(SOC)}{\eta_{conv}}}}{2} \quad (11)$$

According to physical meaning, we choose the "+" sign corresponding to the higher voltage solution when the discriminant is positive. When the discriminant is negative, it indicates that the battery cannot provide the required power P_0 .

3.2.2 Pure Differential Equation System

Substituting Equation (11) into Equation (8) gives the current expression:

$$I(t) = \frac{P_0}{\eta_{conv} V_t(SOC, V_{p1}, V_{p2})} \quad (12)$$

Thus, we can transform the original DAE system into a standard ordinary differential equation (ODE) system:

$$\begin{cases} \frac{dSOC}{dt} = -\frac{\eta P_0}{C_n \cdot SOH \cdot \eta_{conv} V_t(SOC, V_{p1}, V_{p2})} \\ \frac{dV_{p1}}{dt} = \frac{P_0}{C_1 \eta_{conv} V_t(SOC, V_{p1}, V_{p2})} - \frac{V_{p1}}{R_1 C_1} \\ \frac{dV_{p2}}{dt} = \frac{P_0}{C_2 \eta_{conv} V_t(SOC, V_{p1}, V_{p2})} - \frac{V_{p2}}{R_2 C_2} \end{cases} \quad (13)$$

Where $V_t(\cdot)$ is defined by Equation (11). The state variables of the system are $\mathbf{x} = [SOC, V_{p1}, V_{p2}]^T$.

3.2.3 Numerical Solution Algorithm

To solve the ODE system (13), we use two time-discretization methods: the explicit Euler method (first-order accuracy) and the classic Runge-Kutta method of order 4 (RK4, fourth-order accuracy). The Euler method is simple but has limited accuracy, while RK4 has higher accuracy but requires more computational effort.

Explicit Euler Method By discretizing the time domain: $t_k = k\Delta t, k = 0, 1, 2, \dots$, the state update formula of the explicit Euler method is:

$$\begin{cases} SOC_{k+1} = SOC_k - \frac{\eta P_0 \Delta t}{C_n \cdot SOH \cdot \eta_{conv} V_{t,k}} \\ V_{p1,k+1} = V_{p1,k} + \Delta t \left(\frac{P_0}{C_1 \eta_{conv} V_{t,k}} - \frac{V_{p1,k}}{R_1 C_1} \right) \\ V_{p2,k+1} = V_{p2,k} + \Delta t \left(\frac{P_0}{C_2 \eta_{conv} V_{t,k}} - \frac{V_{p2,k}}{R_2 C_2} \right) \end{cases} \quad (14)$$

Where $V_{t,k}$ is computed at time t_k using Equation (11).

Runge-Kutta Method of Order 4 (RK4) By defining the right-hand side function $\mathbf{f}(\mathbf{x})$ as in Equation (13), the single-step update formula of RK4 is:

$$\begin{aligned}
 \mathbf{k}_1 &= \Delta t \cdot \mathbf{f}(\mathbf{x}_k), \\
 \mathbf{k}_2 &= \Delta t \cdot \mathbf{f}\left(\mathbf{x}_k + \frac{\mathbf{k}_1}{2}\right), \\
 \mathbf{k}_3 &= \Delta t \cdot \mathbf{f}\left(\mathbf{x}_k + \frac{\mathbf{k}_2}{2}\right), \\
 \mathbf{k}_4 &= \Delta t \cdot \mathbf{f}(\mathbf{x}_k + \mathbf{k}_3), \\
 \mathbf{x}_{k+1} &= \mathbf{x}_k + \frac{1}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4)
 \end{aligned} \tag{15}$$

3.3 Power Consumption Model

Given that the power demand in real-world scenarios is not constant, we account for time-varying power by coupling a power consumption model with the battery model. The total power consumption $P(t)$ of a smartphone at time t can be expressed as the linear superposition of contributions from all active hardware components:

$$P(t) = P_{\text{basic}} + \sum_{i \in \mathcal{C}} \sum_{k \in \mathcal{V}_i} [c_{i,k}(t) \times \alpha_{i,k}] \tag{16}$$

- P_{basic} denotes the baseline power consumption when the device is active, covering essential operating system services and minimal hardware overhead.
- $\mathcal{C}$ represents the set of all hardware components: $\mathcal{C} = \{\text{CPU, Screen, RAM, etc.}\}$.
- $\mathcal{V}_i$ is the set of system variables associated with component i .
- $\alpha_{i,k}$ is the power correlation coefficient for the k -th system variable of component i .
- $c_{i,k}(t)$ denotes the value of the k -th system variable of component i at time t .

This modular modeling approach allows the contribution of each component to be independently analyzed, calibrated, and updated. Specific parameter values are listed in Table 3¹⁵, where the 5G data are derived by calibrating the original 3G measurements against contemporary benchmarks¹⁷.

Table 3: Power Consumption Coefficients of Hardware Components

Hardware Component	System Variable	Coefficient c (mW)	Hardware Component	System Variable	Coefficient c (mW)
CPU	util	15.936	Phone	callState	753.0
Screen	brightness	4.500	Phone	call_Rssi	90.0
Memory	util	0.156	GPS	GPS_On	9.0
Flash	readSpeed	0.012	Audio	volume	10.47
Flash	writeSpeed	0.021	Audio	headset	66.0
WiFi	wifi_On	141.66	5G	5G_Rssi	3.0
WiFi	wifi_Rssi	0.12	5G	5G_On	1148.4
WiFi	linkSpeed	1272.0	Bluetooth	blue_On	185.4
Camera	camera_On	2218.2	System	System_On	716.4

3.4 Results and Analysis

3.4.1 Battery Discharge Characteristics in Typical Usage Scenarios

By numerically solving the continuous-time battery model and integrating it with the power consumption model, SOC versus time curves under various real-world usage scenarios were obtained.

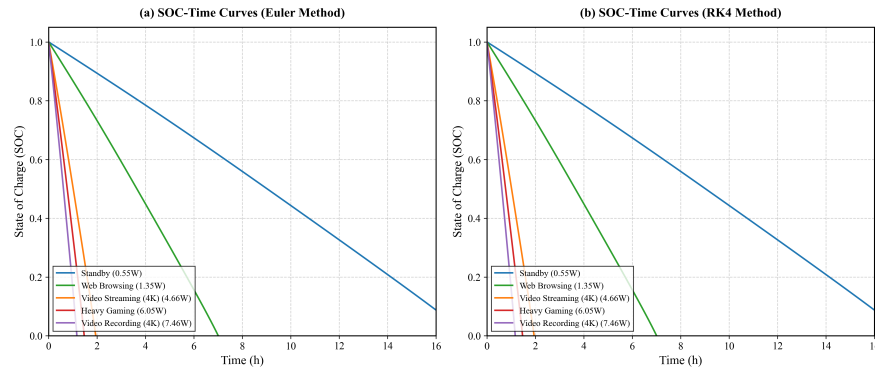


Figure 4: Battery discharge curves under five typical usage scenarios.

Figure 4 presents the SOC-t curves for five typical scenarios, with the following key characteristics:

- Standby Mode:** The curve exhibits the slowest decline, indicating a battery life of approximately 19 hours with an SOC depletion rate of 5.2% per hour.
- Web Browsing:** Representing a moderate power consumption scenario, the discharge time is about 7 hours, corresponding to an SOC depletion rate of 14.3% per hour.
- 4K Video Streaming:** A high power consumption scenario resulting in a discharge time of approximately 2 hours.
- Heavy Gaming:** A high power consumption scenario with a discharge time of about 1.5 hours.

- e) **Video Conferencing:** A sustained high-power scenario leading to a discharge time of about 1.2 hours.

3.4.2 Battery Performance Under a Realistic Daily Usage Pattern

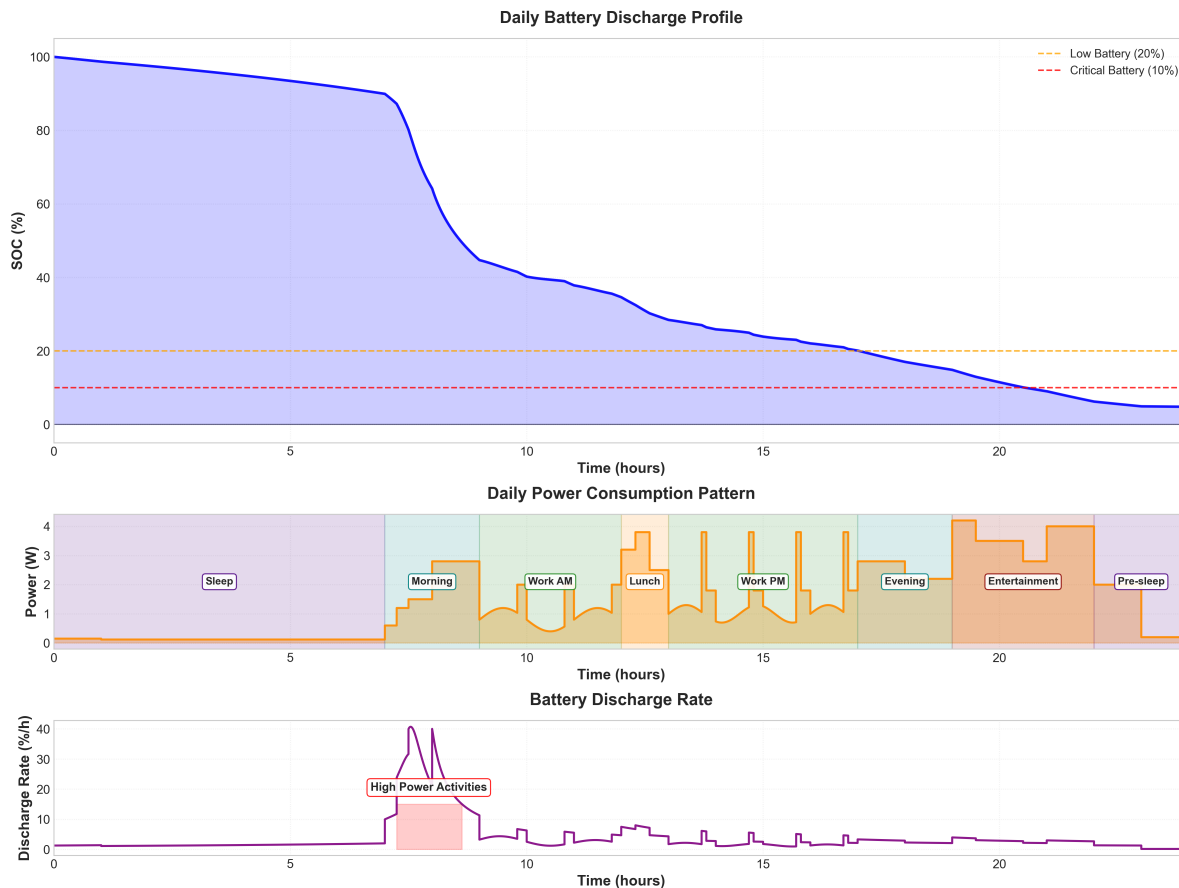


Figure 5: Battery discharge curve under a simulated realistic daily usage pattern.

A 24-hour usage pattern for a typical user, from 7:00 AM to 7:00 AM the next day, was simulated. The results are shown in Figure 5.

Key Findings:

- 1) **Time to Battery Depletion:** 23.5 hours (less than 24 hours), indicating that under the simulated usage pattern, the smartphone requires daily charging. This result aligns with the common experience of most users.
- 2) **Usage Pattern Distribution:**
 - **Sleep Period (7 hours):** Power consumption ranged between 0.12-0.15 W, with an SOC decrease of only about 1%, demonstrating highly effective power management in deep sleep state.

- **Work Period (9 hours):** Average power consumption was 1.5 W, leading to an SOC decrease of approximately 40%. Activities included intermittent communication, information browsing, and short breaks.
- **Entertainment Period (4 hours):** Average power consumption peaked at 3.5 W, causing an SOC decrease of about 45%. This period involved high-power activities such as gaming and video streaming.

3) High Power Consumption Intervals:

- **Lunch Break (12:00-13:00):** SOC decreased by 8%, primarily due to short video viewing and casual gaming.
 - **Evening Entertainment (19:00-22:00):** SOC decreased significantly by 35%, dominated by extended gaming and video streaming sessions.
- 4) **Variation in Discharge Rate:** The maximum discharge rate of 15% per hour occurred during the evening gaming session. In contrast, the discharge rate during sleep dropped to 0.14% per hour—a difference exceeding two orders of magnitude.

3.4.3 Comparative Analysis of Power Consumption Across Scenarios

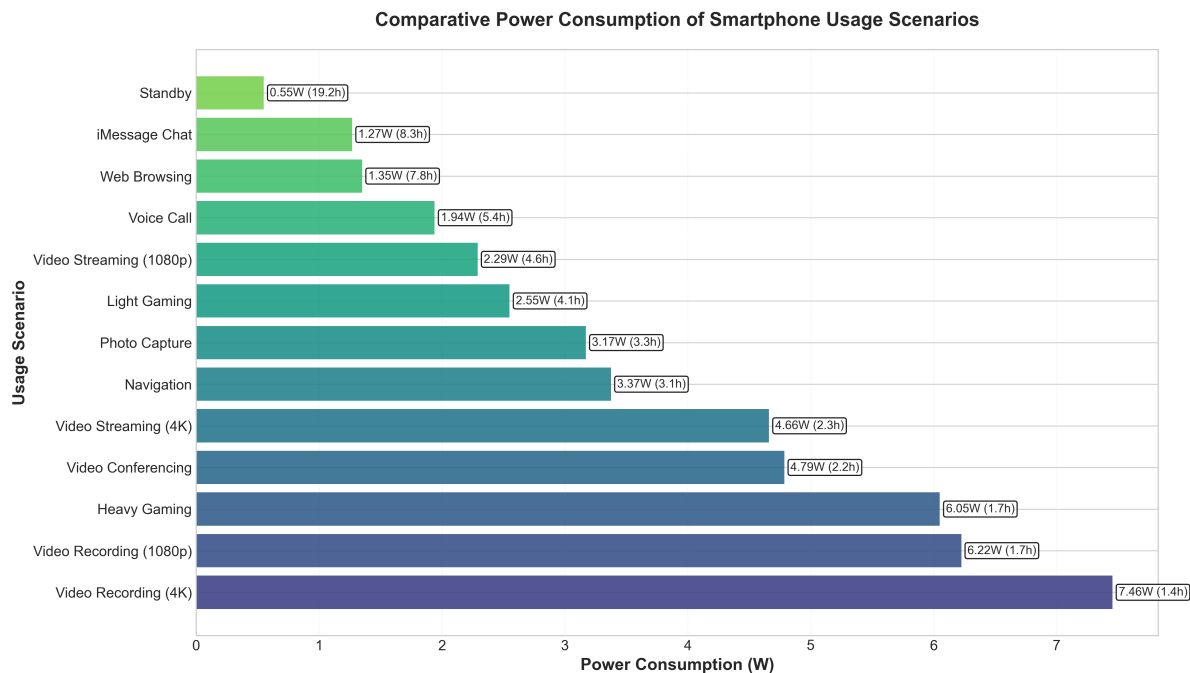


Figure 6: Comparison of power consumption across different usage scenarios.

Figure 6 provides a comparison of power consumption for various scenarios, with the principal findings summarized below:

- a) **Highest Power Consumption Activities:** Video recording (1080p) > heavy gaming > video conferencing > video streaming. The high power demand in gaming and recording is primarily attributed to sustained high GPU load.

- b) **Medium Power Consumption Activities:** Navigation > photography > 1080p video streaming. The significant power draw during navigation results from continuous screen operation at high brightness and sustained GPS activity.
- c) **Low Power Consumption Activities:** Voice calls > Web browsing > iMessage communication. These activities are characterized by intermittent CPU and network usage.
- d) **Standby Power Consumption:** This represents the lowest power state, consuming merely 4% of the power required by intensive gaming. This indicates that power management in idle states is highly optimized in modern smartphones.

Power Composition Analysis: For the intensive gaming scenario, the power consumption is primarily distributed as follows: screen (30%), GPU (30%), CPU (25%), and other components (10%). Reducing screen brightness can significantly decrease the total power consumption.

4 Problem 2: Discharge Time Prediction Under Realistic Conditions

4.1 Problem Restatement and Mathematical Formulation

The primary objective of this section is to predict the *discharge time* (time-to-empty, TTE) given an arbitrary initial state and a specific usage scenario. Our continuous-time model accounts for the nonlinear voltage drop induced by internal resistance and polarization effects under a constant-power load.

We define the discharge time as the moment the system state first meets a termination criterion. Let the system state be $\mathbf{x}(t) = [\text{SOC}(t), V_{p1}(t), V_{p2}(t)]^T$ and the load power be $P(t)$. Its dynamics are governed by the differential equations derived in Section 1:

$$\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}, P(t), \theta) \quad (17)$$

where θ represents the battery parameters (R_0, R_1, C_1 , etc.).

Termination of a single discharge cycle occurs when the state of charge depletes to zero or the terminal voltage falls below a cutoff threshold $V_{\min}$. The terminal voltage $V_t(t)$ is an implicit function of the state and load:

$$V_t(t) = V_{OC}(\text{SOC}(t)) - V_{p1}(t) - V_{p2}(t) - I(t)R_0(\text{SOC}(t)). \quad (18)$$

Under the constant-power constraint $P(t) = P_{\text{set}}$, the current $I(t)$ couples with the voltage via $I(t) = P_{\text{set}}/V_t(t)$. Solving this algebraic loop yields an explicit voltage function $g(\mathbf{x}, P_{\text{set}})$:

$$V_t(t) = \frac{V_{OC} - V_p + \sqrt{(V_{OC} - V_p)^2 - 4P_{\text{set}}R_0(\text{SOC})}}{2} \quad (19)$$

where $V_p = V_{p1} + V_{p2}$.

Consequently, the discharge time prediction problem is formulated as finding the minimum time interval $\mathcal{T}$:

$$\mathcal{T}(\text{SOC}_0, P_{\text{set}}) = \inf\{t > 0 \mid V_t(t) \leq V_{\min} \vee \text{SOC}(t) \leq 0\}. \quad (20)$$

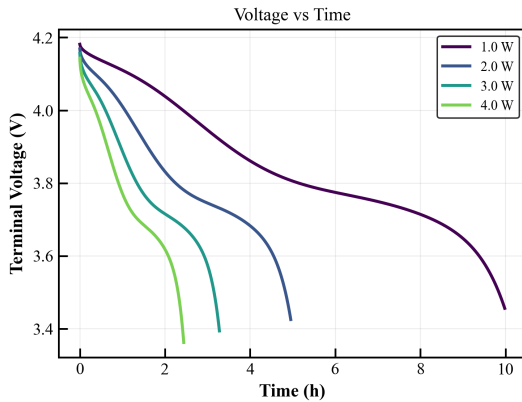
4.2 Numerical Solution and Dynamic Analysis

Due to the highly nonlinear OCV-SOC relationship and polarization dynamics, an analytical solution for $\mathcal{T}$ is intractable. We employ a Runge-Kutta (RK45) adaptive time-step integration method to solve this ODE system until the event function $E(t) = V_t(t) - V_{\min}$ crosses zero. The results are as follows.

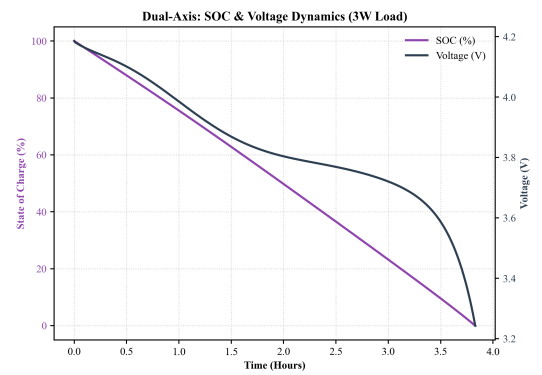
4.2.1 Voltage Collapse Under High Load

A key insight from our model is the distinction between “energy depletion” ($\text{SOC} = 0$) and “voltage collapse” (voltage $< V_{\min}$). As shown in **Fig. 7a**, under low power loads (e.g., 0.5 W), the battery discharges completely until SOC nears zero. However, under high power loads, the significant internal resistance drop (IR_0) causes the terminal voltage to drop to a lower level compared to low-power scenarios, readily hitting the 3.0 V cutoff threshold early and leaving “stranded energy” within the battery.

The dual dynamics of SOC and voltage are further illustrated in **Fig. 7b**. While SOC declines approximately linearly (resembling an ideal energy reservoir), the voltage exhibits nonlinear curvature due to the diffusion processes modeled by the parallel RC branches.



(a) Terminal voltage trajectories under different constant-power loads.



(b) Synchronous evolution of SOC and terminal voltage (Load = 3 W).

4.3 Comprehensive Discharge Time Prediction Landscape

We performed a global analysis across the operational domain, varying the initial SOC from 10% to 100% and the constant power load from 0.1 W to 3.0 W.

The results are visualized in the discharge time (TTE) landscape plot (**Fig. 8**). This contour plot serves as a lookup table for our model, displaying the predicted remaining battery time across different initial SOC values and power levels.

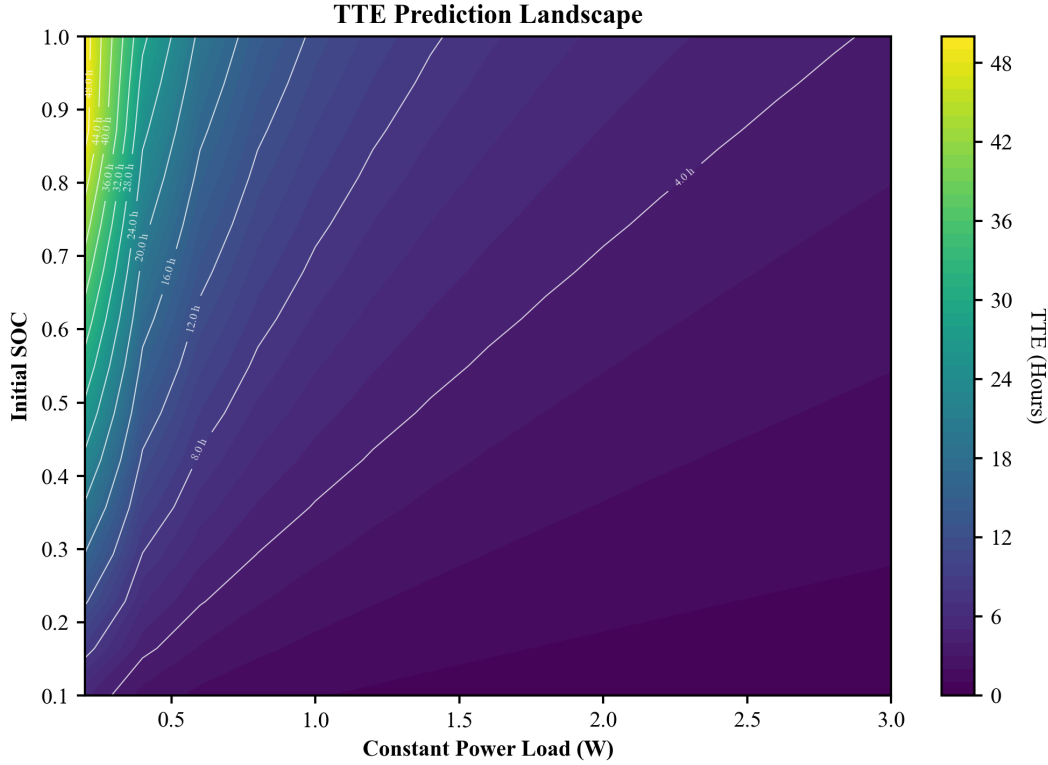


Figure 8: Global discharge time (TTE) prediction landscape

4.3.1 Analysis of the TTE Prediction Landscape

The global discharge time (TTE) prediction landscape in Fig. 8 reveals the coupled effects of initial SOC and constant load power on battery discharge duration. Initial SOC is the dominant factor determining TTE, with discharge time increasing approximately linearly with rising SOC under a fixed power load.

Conversely, load power exerts a nonlinear suppressing effect on TTE, causing it to decrease rapidly as power increases—this trend is particularly pronounced in high-SOC regions. The oblique and increasingly dense contour lines further indicate a nonlinear coupling between initial SOC and load power.

This coupling implies that increasing the initial SOC yields greater marginal benefits for extending TTE in high-power scenarios (e.g., gaming) compared to low-power scenarios (e.g., standby).

Analysis of the boundary behavior identifies the extreme values of TTE: the longest discharge time (50 hours) occurs at high initial SOC combined with low load power, while the shortest discharge time (<2 hours) is observed at low initial SOC under high load power.

These findings provide a quantitative basis for optimizing smartphone battery life management and also validate the simulation accuracy of the second-order RC model.

5 Problem 3: Sensitivity Analysis and Robustness Assessment

To evaluate the reliability of the proposed Second-Order RC model under parameter perturbations and extreme operating conditions, we conducted a comprehensive sensitivity analysis. This section investigates the non-linear coupling effects between load power and battery capacity, quantifies the uncertainty arising from manufacturing tolerances using Monte Carlo simulations, and validates the model's dynamic response under realistic scenarios.

5.1 Non-linear Impact of Power Load

The sensitivity of the Time-to-Empty (TTE) prediction to variations in power load is illustrated in Figure 9.

Power Load Sensitivity Analysis for TTE Prediction

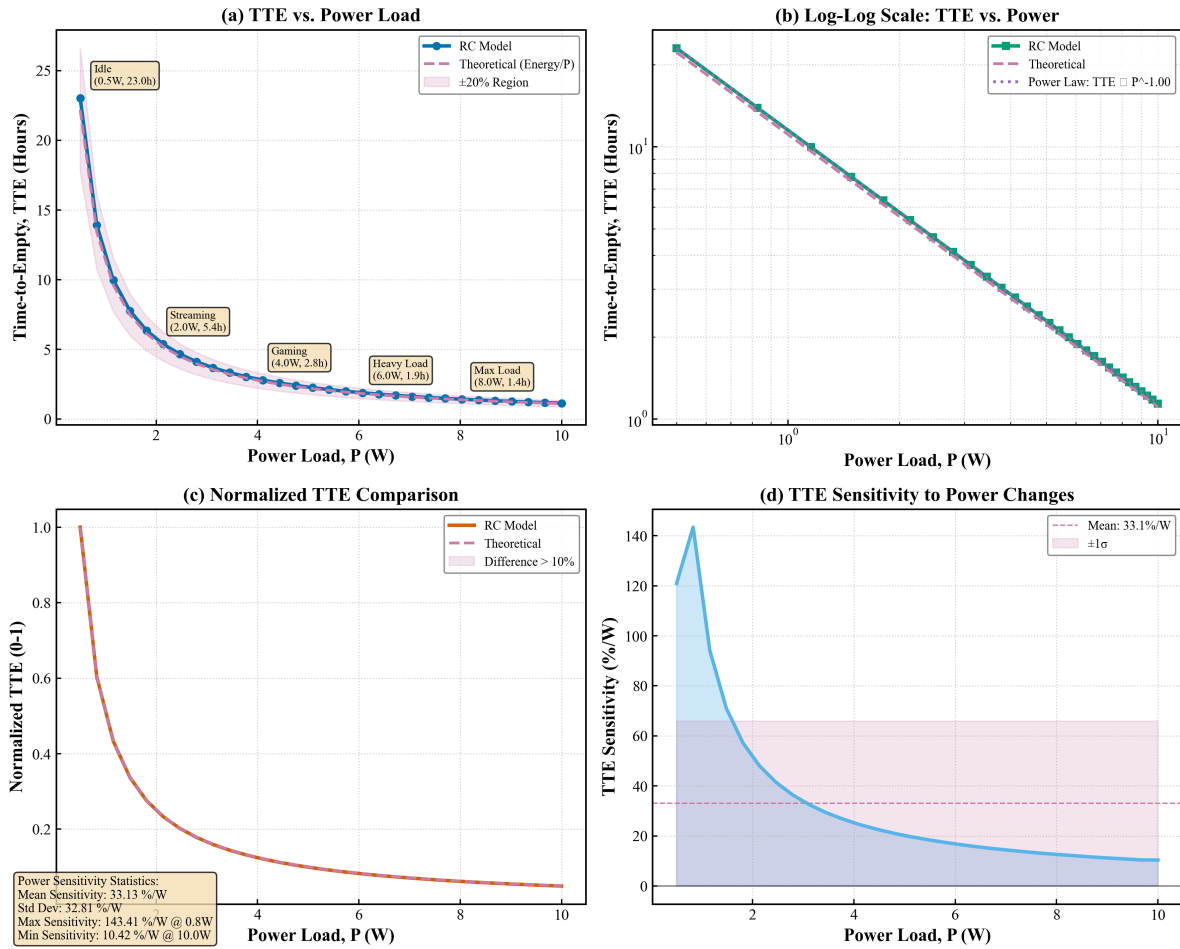


Figure 9: Sensitivity analysis of TTE with respect to Power Load (P). (a) Comparison between model prediction and theoretical energy-based estimation; (b) Log-log scale relationship; (c) Normalized TTE indicating efficiency loss; (d) Gradient analysis showing sensitivity variations.

As shown in Figure 9(a), TTE generally follows the inverse power law ($TTE \propto P^{-1}$). However, Figure 9(d) reveals that the system is extremely sensitive in the low-power region (Sensitivity > 100%/W at 0.8W), implying that background optimization yields the highest marginal gain.

5.2 Capacity-Power Coupling Trade-off

We further explored the interplay between battery capacity (C) and power load, as presented in Figure 10.

Battery Capacity Sensitivity Analysis for TTE Prediction

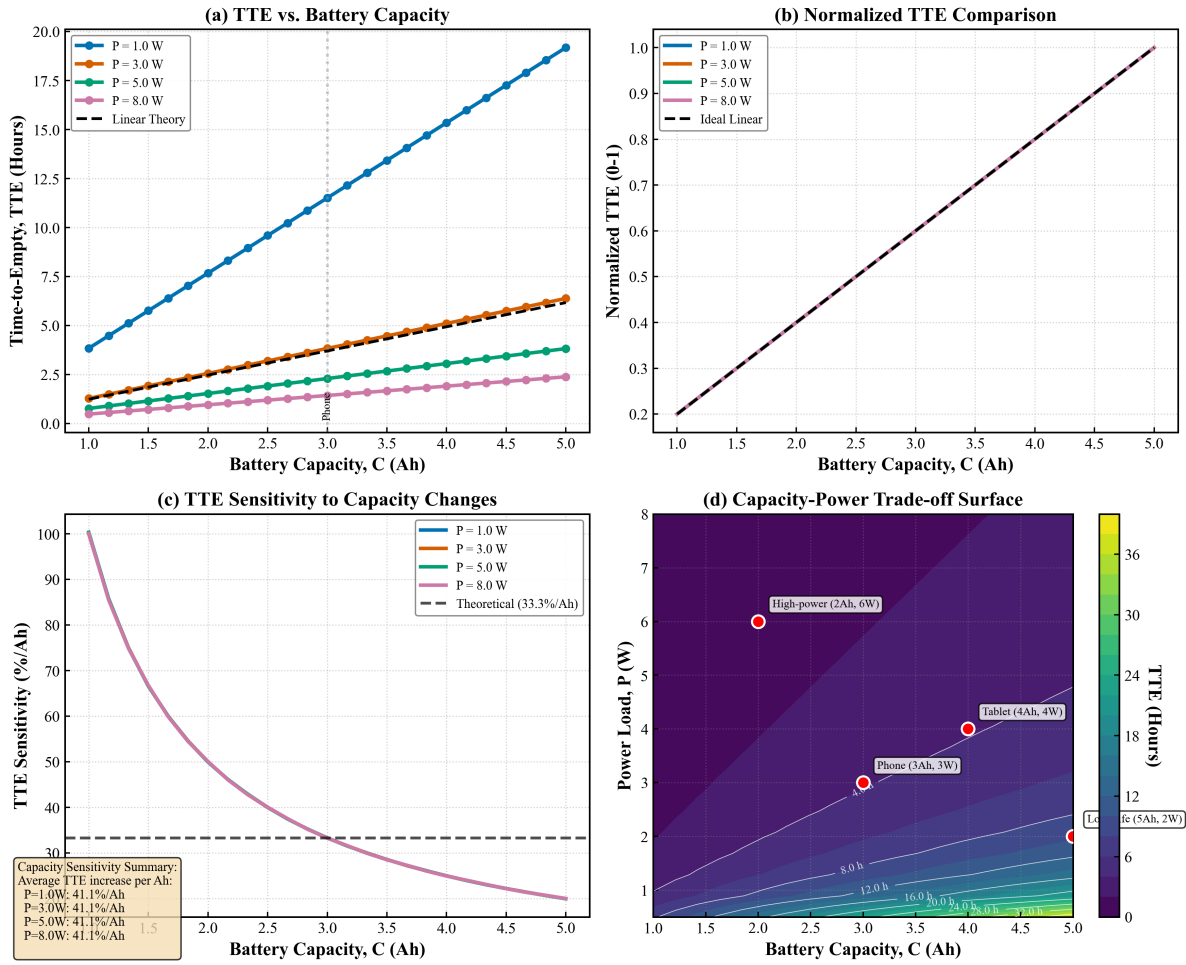


Figure 10: Sensitivity analysis of TTE with respect to Battery Capacity (C)

Collectively, these figures demonstrate that Time-to-Empty (TTE) increases linearly with battery capacity across all power loads, though the growth rate is significantly dampened at higher power levels. Normalized TTE analysis confirms this linear relationship is universal, while sensitivity curves reveal that capacity improvements yield greater relative gains in TTE for smaller-capacity batteries. The capacity-power trade-off surface further illustrates that longer TTE is achieved by combining higher capacity with lower power, offering a clear quantitative basis for optimizing battery-life management.

5.3 Uncertainty Quantification via Monte Carlo Simulation

To account for the stochastic nature of battery manufacturing (e.g., variations in R_0 , C , and aging effects), we performed a Monte Carlo simulation with $N = 500$ iterations. The parameters were sampled from normal distributions centered at nominal values with a 5-10% standard deviation.

Monte Carlo Uncertainty Analysis of TTE Prediction

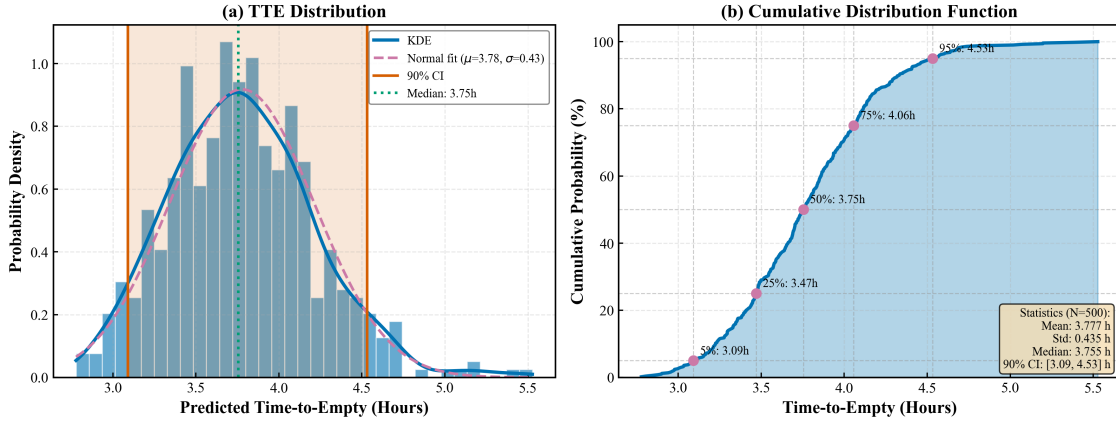


Figure 11: Uncertainty quantification. (a) Histogram and KDE of predicted TTE; (b) Cumulative Distribution Function (CDF) showing confidence intervals.

The Monte Carlo simulation ($N = 500$) results are presented in Figure 11. The TTE distribution is approximately normal with a mean of **3.75 hours** and a standard deviation of **0.43 hours**. The 90% confidence interval spans from **3.09 h to 4.53 h**. The distribution exhibits a slight right skew, and the notable gap between the 95th percentile and the mean underscores the necessity of using conservative estimates in reliable power management systems.

Finally, we validated the model's transient behavior under dynamic loading conditions, which represent real-world usage more accurately than constant-power tests.

Figure 12(a) highlights the voltage sags during the "Gaming" scenario (red curve). Unlike the smooth decline in "Streaming," the high-frequency load variations cause instantaneous voltage drops. If these drops hit the cutoff voltage, the device may shut down even if the State of Charge (SOC) is above 0%, a phenomenon known as "voltage collapse."

Figure 13 further confirms the model's fidelity. The voltage relaxation observed after power steps (Figure 13(b)) is perfectly captured by the two RC time constants (τ_1, τ_2), demonstrating that our model correctly accounts for the diffusion and charge-transfer delays that are critical for accurate TTE prediction in dynamic environments.

Scenario Analysis: Discharge Behavior Under Different Power Profiles

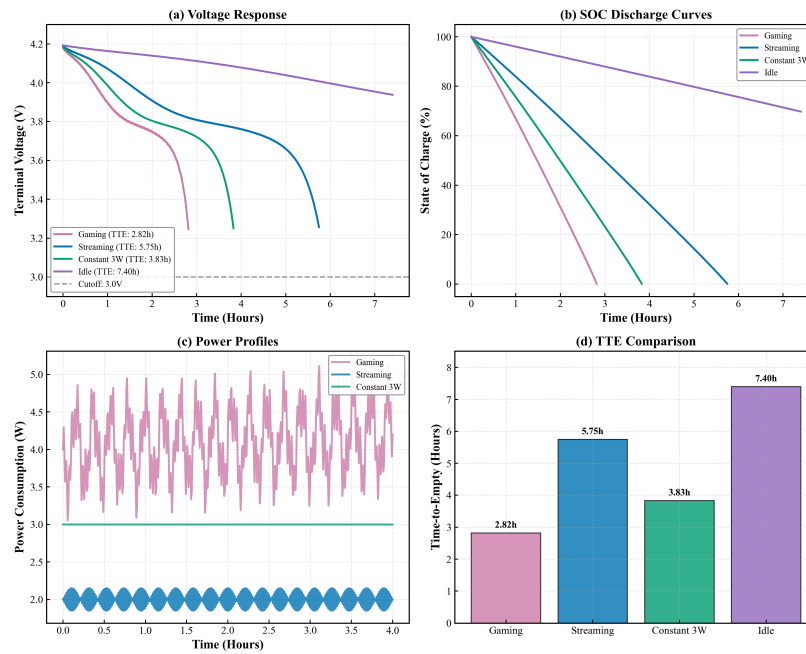


Figure 12: Simulation of diverse usage scenarios. (a) Terminal voltage response; (c) Dynamic power profiles including Gaming, Streaming, and Idle.

Model Validation: 2nd-Order RC Dynamic Response to Step Power

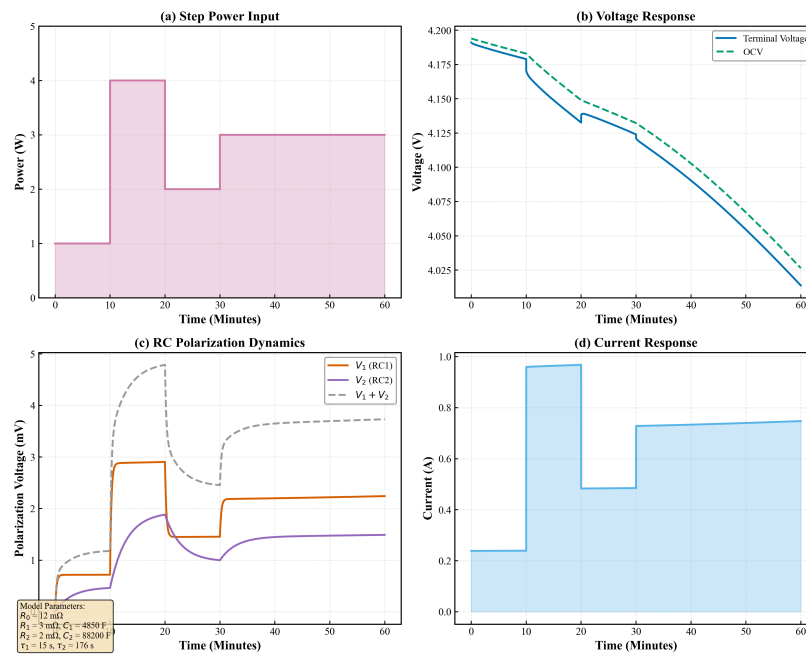


Figure 13: Model validation via Step Response

6 Problem 4: Practical Implementation and Model Extensions

Based on the quantitative insights derived from the sensitivity analysis and uncertainty quantification, we translate our theoretical findings into actionable strategies for end-users and operating systems. Furthermore, we analytically extend the framework to account for thermal dynamics and aging effects.

6.1 User Guidelines: Exploiting the High-Sensitivity Region

Our Power Load Sensitivity Analysis Fig.10 reveals a critical insight: **TTE sensitivity (%/W) is non-linear and peaks at low power loads.** Specifically, the sensitivity at the "Idle" state (0.8 W) is over 140%/W, whereas it drops to $\sim 10\%/W$ at "Max Load".

- **High Marginal Utility of Low-Power Optimization:** Users should prioritize reducing "background drain" rather than active heavy tasks. Dimming the screen brightness or switching from 5G to 4G (saving ~ 0.5 W) while browsing yields a disproportionately larger extension in battery life compared to lowering graphics settings during gaming.
- **The "Kill-Switch" Rule:** Since the derivative $|\frac{\partial TTE}{\partial P}|$ is maximized near zero load, disabling unused background sensors (GPS, Bluetooth scanning) provides the most efficient TTE gain per unit of user inconvenience.

6.2 OS Strategy: Probabilistic Power Management

Because of the risk of voltage collapse under high-power operation, we recommend the following two strategies:

Confidence-Based UI Display: Instead of displaying the mean TTE, the OS should display the **Conservative Prediction (90% Confidence Level)**. This prevents unexpected shutdowns, aligning user expectations with the worst-case physical reality.

Proactive Voltage Governor: Our 2nd-Order RC model predicts "Voltage Collapse" before SOC depletion. The OS should implement a *Preemptive Throttling* mechanism: when V_{term} approaches the cutoff threshold $V_{cutoff} + \delta$, the system should forcibly throttle CPU frequency to reduce current I , thereby reducing the ohmic voltage drop IR_0 and instantly recovering terminal voltage to prolong runtime.

6.3 Model Extension: Analytical Derivation for Temperature and Aging

Due to space constraints, we extend our model via analytical derivation rather than numerical simulation. We introduce correction factors to the core TTE equation.

1. Temperature Correction (Arrhenius & Polarization Model): Temperature affects both the Ohmic resistance (R_0) and polarization dynamics. Based on the Arrhenius equation, the total effective resistance $R_{tot}(T)$ is:

$$R_{tot}(T) = \underbrace{R_{pol}}_{\text{Polarization}} + R_{ref} \cdot \exp \left[\frac{E_a}{k_B} \left(\frac{1}{T} - \frac{1}{T_{ref}} \right) \right] \quad (21)$$

Where $R_{pol} = R_1 + R_2$ represents the steady-state polarization equivalent. The corrected TTE accounts for premature voltage cutoff:

$$TTE_{temp} \approx \underbrace{\frac{C_{nom} \cdot \bar{V}_{oc}}{P_{load}}}_{\text{Ideal TTE}} \cdot \left(1 - \frac{P_{load} \cdot R_{tot}(T)}{V_{min} \cdot \kappa_{ocv}} \right) \quad (22)$$

Implication: In cold environments, $R_{tot}(T)$ spikes. The term in parentheses (Derating Factor) decreases significantly, reflecting that the battery hits V_{min} early due to large internal voltage drops (IR_{tot}), stranding usable energy.

2. Aging Correction (SOH Model): Aging manifests as Capacity Fade and Resistance Growth. We define State of Health (SOH) influencing both parameters:

$$C_{aged} = C_{nom} \cdot SOH, \quad R_{aged} = R_{nom} \cdot (1 + \gamma(1 - SOH)) \quad (23)$$

Where γ is the impedance growth factor affecting both ohmic and polarization parts. The generalized TTE for an aged device is:

$$TTE_{aged} \approx SOH \cdot \frac{E_{nom}}{P_{load}} \cdot \left(1 - \frac{P_{load} \cdot R_{aged}}{V_{min} \cdot \kappa_{ocv}} \right) \quad (24)$$

Conclusion: Aging reduces TTE via a "Double Penalty": linearly by scaling down capacity (SOH), and non-linearly by increasing the voltage collapse penalty (R_{aged}), causing the device to shut down at higher real SOC's.

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Report on the Use of Artificial Intelligence

In the preparation of this paper, we utilized Large Language Models (LLMs) to enhance the efficiency of coding and the clarity of our writing. The specific usage and our validation process are detailed below:

- **Google Gemini (Model: 3 PRO):**

- **Usage:** Used for critical review and refinement of the manuscript text. Specifically, we employed Gemini to polish the Abstract and Introduction to ensure the academic tone was formal and coherent. We also used it to check the logical flow of our arguments regarding the “Voltage Collapse” phenomenon.
- **Human Oversight:** All suggestions made by the AI were manually reviewed. We accepted changes related to grammar and flow but rejected any modifications that altered the physical interpretation of our model.

- **DeepSeek (Model: DeepSeek-V3/R1):**

- **Code Assistance:** Used to generate auxiliary code snippets for solving the Differential-Algebraic Equations (DAE) associated with the 2nd-Order RC model. DeepSeek assisted in debugging syntax errors in our Python/MATLAB scripts and optimizing the visualization of the Monte Carlo simulation results.
- **Translation Assistance:** Used to translate specific technical descriptions from our native language to English to ensure terminological accuracy.
- **Human Oversight:** The mathematical logic and governing equations in the code were defined entirely by the authors. We validated the AI-generated code by comparing the simulation outputs against theoretical expectations. The translated text was cross-checked to ensure it accurately reflected the original technical intent.

Declaration: The conceptualization of the Second-Order RC model, the derivation of the coupled power dynamics, the data analysis, and the final conclusions are the original work of the authors.